

02.2 Extraction Codebook

Data Extraction Protocol

1. Overview

The extraction framework comprises 45 fields organised into 10 conceptual blocks. Data were extracted at the tool level: studies evaluating multiple distinct responsible gambling (RG) tools yielded multiple extraction records. The framework captured evidence across experimental, observational, qualitative, and policy-level designs while preserving the distinctions in what each could evidence.

2. Extraction Framework

Table 02.2.1 lists the 45 formal fields; a sortable version is in 02.1 Field Inventory. In the deposited dataset, some fields were stored as compound columns (sub-components delimited within a single cell) and Study_Sample was split across three columns. Several sub-fields used controlled vocabularies (introduced from v8), with codes in the Controlled_Vocabularies sheet of [OSF: 03_1_Taxonomy_Dataset.xlsx]. Storage conventions are in Section 5.8 and do not alter the field count.

Table 02.2.1

Data Extraction Fields by Conceptual Block

Field	Stored in column	Definition	Values / format
Block 1: Identification			
Paper_ID	Paper_ID	Unique paper identifier. One per source paper; shared across all extractions from the same paper.	Format: [Year]_[FirstAuthor]_[KeyTopic]
Study_Category	Study_Category	Classification of the study's design and evidence contribution. Governs the language mode used in results reporting.	EXP [EFFECTIVENESS] / OBS [QUANTITATIVE – REACH] / QUAL [REACH – IMPLEMENTATION] / POL [POLICY – IMPLEMENTATION]
Study_ID	Study_ID	Unique extraction identifier. One per extracted tool record; a paper evaluating multiple tools yields multiple Study_IDs.	Format: [Category]_[Year]_[Author]_[ToolCode]
Block 2: Bibliographic			
Authors	Authors	All authors of the source paper, in citation order.	Last, Initials (comma-separated; & before final author)
Year	Year	Publication year of the source paper.	YYYY
Title	Title	Full article title.	Sentence case
Journal	Journal	Publication venue: full journal name, not abbreviation.	Include subtitle where one exists
DOI	DOI	Digital object identifier linking to the source paper.	https://doi.org/...

Field	Stored in column	Definition	Values / format
Block 3: Study Design			
Study_Design	Study_Design	Research design with key features. Stored as controlled-vocabulary code (from v8) followed by free-text bracketed specifications.	Code: RCT / LAB-EXP / QUASI-EXP / OBSERVATIONAL / CROSS-SECTIONAL / QUALITATIVE / MIXED-METHODS / AUDIT / SINGLE-ARM / DELPHI / OTHER. Format: CODE (free text: arm counts, assignment, timepoints, setting). Single code only.
Data_Source	Data_Source	Type of data used and how it was collected. Stored as controlled-vocabulary code(s) followed by free-text descriptors.	Codes: SELF-REPORT / BEHAVIOURAL-LAB / OPERATOR-LOGS / QUAL-INTERVIEW / OBSERVATIONAL-AUDIT / SECONDARY-SURVEY / SECONDARY-ADMIN / OTHER. Combinable with +. Format: CODE [+ CODE] (free text).
Study_Sample	Sample_N + Sample_Description + Sample_Severity	Description of the study's sample (who was studied). Distinct from Tool_Target_Tier (who the tool is designed for). Captured across three columns in the deposited dataset; counts as one formal field (see Section 5.8).	Begin with risk level in caps; include N, demographics, recruitment, screening
Block 4: Setting			
Country_Jurisdiction (Country + Regulatory_Context)	Country_Jurisdiction	Geographic and regulatory context. Compound field containing Country and Regulatory_Context.	Country + region/state where sub-national variation is relevant; licensing authority + key legislation or industry code
Game_Type (Platform_Type + Money_Type)	Game_Type	Gambling product type and money type. Compound field containing Platform_Type and Money_Type.	Platform_Type (EGM/Slots/VLT, Sports/Racing, Casino/Table, Poker, Lottery/Bingo, General online, Mixed); Money_Type (real-money / simulated)
Platform_Setting	Platform_Setting	Platform environment where gambling takes place.	Online (operator named where relevant) / Land-based / Laboratory / Mixed / Non-specific
Block 5: Tool Identity			
Tool_Code	Tool_Code	Standardised tool code capturing mechanism of action. Combined codes permitted for integrated multi-component tools (e.g., FEED-PERSONAL + MSG-POP).	See 02.3 Tool Codes and Domains for full code definitions and bundling rules

Field	Stored in column	Definition	Values / format
Tool_Name	Tool_Name	Descriptive name of the tool as deployed or evaluated.	Use dashes, not colons
Tool_Description	Tool_Description	What the tool IS and DOES: core function and mechanism of action.	Prose, 2–4 sentences; omit study methodology and delivery parameters
Tool_Content	Tool_Content	Verbatim content or components of the tool where available (e.g., exact message wording, interface elements, intervention script).	Verbatim in quotes; [bracketed paraphrase] where exact wording unavailable
Tool_Target_Tier	Tool_Target_Tier	Risk tier the tool is designed for. Captures targeting logic at tool level. Distinct from Study_Sample (who was actually studied). N/A where the record has no human participant sample (policy audits, document analyses, simulation-only studies).	ALL / HIGH / MOD / LOW / N/A
Block 6: Delivery			
Delivery_Mechanism (Modality + Timing + Trigger + Frequency + Duration)	Delivery_Mechanism	How, when, and how often the tool is delivered. Compound field containing Modality, Timing, Trigger, Frequency, and Duration. Each sub-field uses controlled vocabulary codes (from v8) followed by free-text descriptors.	Modality (PLATFORM-INTEGRATED / POP-UP / ONLINE-MODULE / VIDEO / STATIC-DISPLAY / SIGNAGE / SMS / EMAIL / LETTER / TELEPHONE / FACE-TO-FACE / WORKBOOK / OPERATOR-BACKEND / OTHER; combinable with +); Timing (PRE-SESSION / DURING-PLAY / POST-SESSION / BETWEEN-SESSIONS / ON-DEMAND / SCHEDULED / CONTINUOUS / N/A; combinable with +); Trigger (AUTOMATIC / USER-INITIATED / OPERATOR-INITIATED / TIME-BASED / SPEND-THRESHOLD / BEHAVIOUR-THRESHOLD / EVENT-TRIGGERED / N/A; combinable with +); Frequency (ONE-TIME / REPEATED / PERIODIC / CONTINUOUS / ON-DEMAND / N/A; single code only); Duration (free text)
User_Interaction (Control + Engagement + Personalisation)	User_Interaction	User engagement requirements. Compound field containing Control, Engagement, and Personalisation. Each sub-field uses controlled vocabulary codes (from v8) followed by free-text descriptors.	Control (MANDATORY / VOLUNTARY / MANDATORY+VOLUNTARY / N/A; PASSIVE is not a Control value); Engagement (PASSIVE / ACKNOWLEDGE / VIEW-

Field	Stored in column	Definition	Values / format
			REQUIRED / INPUT-REQUIRED / INTERACTIVE / N/A; two compound combinations permitted: PASSIVE+INPUT, VIEW+INPUT); Personalisation (NONE / USER-SET / BEHAVIOURAL / INDIVIDUALISED / N/A; one compound combination permitted: USER-SET+BEHAVIOURAL)
Block 7: Measurement			
Primary_Outcomes	Primary_Outcomes	Main dependent variables. Outcome types span behavioural, psychosocial, engagement, and acceptability.	Outcome construct + measurement; one per line where multiple
Secondary_Outcomes	Secondary_Outcomes	Additional measures including covariates, moderators, and secondary dependent variables.	Covariates and moderators; semicolon-separated
Measurement_Method (Instruments + Timepoints + Procedure)	Measurement_Method	Assessment procedures. Compound field containing Instruments, Timepoints, and Procedure.	Instruments (validated instruments named, e.g., PGSI, DASS-21); Timepoints (baseline + follow-up intervals); Procedure (administration mode + setting)
Analysis	Analysis	Statistical or analytical methods used.	Statistical methods only; exclude substantive interpretation
Block 8: Results			
Main_Findings	Main_Findings	Key results with statistics. Language mode must match Study_Category (causal for EXP; associational for OBS; interpretive for QUAL; descriptive for POL).	Quantitative results first; language mode per Study_Category
Subgroup_Effects	Subgroup_Effects	Differential outcomes by risk level, demographics, gambling format, or other reported moderators.	By moderator: effect. 'Not reported' where absent
Effect_Sizes	Effect_Sizes	Magnitude of effects. Required for EXP records; N/A for OBS, QUAL, and POL.	EXP: d, η^2 , OR, RR with CIs where available. OBS/QUAL/POL: N/A
Acceptability	Acceptability	User engagement and satisfaction data.	Uptake rate + satisfaction + qualitative feedback where available
Block 9: Implementation			
Implementation_Readiness	Implementation_Readiness	Tool readiness level based on	Deployed / Validated / Tested /

Field	Stored in column	Definition	Values / format
diness	diness	evidence context. Describes the tool's readiness, not the study's contribution. Parenthetical qualifier specifies context (e.g., 'Deployed (multi-operator, policy audit)').	Conceptual
Implementation_Notes	Implementation_Notes	Requirements, design levers, and structural barriers for deploying the tool in digital gambling environments. Required for every record regardless of original study context (see Section 5.7).	Online translation requirements; structural barriers; design levers
Evidence_Quality	Evidence_Quality	Five-level quality rating with direction indicator and interpretive statement.	[RATING] (direction). [Interpretive statement]. Strengths: [...]. Limitations: [...]. Ratings: HIGH / MODERATE-HIGH / MODERATE / LOW-MODERATE / LOW. Directions: positive / negative / mixed / null
Block 10: Context			
Behavioural_Psychological_Social	Behavioural_Psychological_Social	Mechanisms and psychological principles underpinning the tool. Content varies by Study_Category (see Table 02.2.3 for study-type-specific interpretations).	Content varies by Study_Category
Risk_Factors_Indicators	Risk_Factors_Indicators	Behavioural markers relevant to tool deployment or targeting. Content varies by Study_Category (see Table 02.2.3).	Content varies by Study_Category
Economic_Behavioural_Economics	Economic_Behavioural_Economics	Financial and behavioural-economic factors. Content varies by Study_Category (see Table 02.2.3).	Content varies by Study_Category

The 45 fields comprise 30 standalone fields and 15 sub-fields nested within five compound fields (Country_Jurisdiction, Game_Type, Delivery_Mechanism, User_Interaction, Measurement_Method); they are stored in 37 columns (35 single-cell or compound columns, with Study_Sample split across three).

Note. RG = responsible gambling. Compound field = single column with delimiter-separated sub-components. The SELF-REPORT code in the Data_Source vocabulary is namespace-distinct from the SELF-REG tool code in 02.3 Tool Codes and Domains. SELF-REPORT identifies how data were collected; SELF-REG identifies a category of self-regulation tools.

3. Study Category Classification

Studies were classified into four categories by one question: did the researchers deliberately manipulate or deliver the RG tool with a comparison condition? Classification governed language mode in results reporting (Section 5.4) and interpretation of the Block 10 context fields (Section 4).

Table 02.2.2
Study Category Definitions and Language Modes

Category	Definition	Language mode	Effect sizes
EXP	Experimental studies with manipulation and comparison condition	Causal: “reduced,” “caused,” “led to”	Required (d, η^2 , OR, RR)
OBS	Quantitative studies of naturally occurring tool use	Associational: “showed,” “was associated with”	N/A
QUAL	Interpretive exploration of experiences and processes	Interpretive: “participants described,” “themes included”	N/A
POL	Regulatory, compliance, or industry-level studies	Descriptive: “X% provided,” “compliance was”	N/A

4. Context Fields by Study Type

The three Block 10 context fields were interpreted differently by study category to capture the distinct contributions of each evidence type. Field names in the deposited dataset preserved the EXP/OBS form; QUAL and POL interpretations were stored in the same cells with an inline prefix ([QUAL: ...] or [POL: ...]).

Table 02.2.3
Context Field Interpretation by Study Category

Field	EXP / OBS	QUAL	POL
Block 10, field 1	Behavioural_Psychological_Social: mechanisms, behavioural targets, psychological principles	User_Experience_Themes: key themes from participant perspectives, emotional responses	Policy_Regulatory_Context: legislation, enforcement, jurisdictional details
Block 10, field 2	Risk_Factors_Indicators: behavioural markers for tool deployment, success/failure indicators	Barriers_Facilitators: factors helping or hindering engagement, motivations, deterrents	Implementation_Compliance : compliance rates, barriers, quality issues
Block 10, field 3	Economic_Behavioural_Economics: financial mechanisms, operator considerations	Design_Implications: actionable insights for tool design, user preferences	Economic_Policy_Implications: costs, resources, industry burden

5. Key Extraction Rules

5.1 Unit of Extraction

Data were extracted at the tool level. A single paper yielded multiple extractions if it:

- Tested different tool types (e.g., pop-up messages and deposit limits)
- Compared meaningfully different versions of a tool (e.g., generic warning vs. personalised feedback)
- Assessed multiple RG features in a compliance audit

A single extraction was used when conditions were variations of the same tool (e.g., dose or frequency). Where a tool combined mechanisms from multiple categories (e.g., FEED-PERSONAL + MSG-POP), combined codes were used per the bundling rules in 02.3 Tool Codes and Domains.

5.2 Self-Contained Records

Each extraction was self-contained. Cross-references (e.g., “see above”) were not permitted. All information needed to interpret a record sat within that record.

5.3 Tool Description Standards

The Tool_Description field captured what the tool IS and DOES: not study methodology, delivery timing, or user requirements. Content focused on:

- Core function of the tool
- Content type or information provided
- Mechanism of action

5.4 Language Mode Enforcement

Causal language (“reduced,” “caused,” “prevented”) was permitted only for EXP studies. For OBS studies, even when analysing tool users, associational language was required: “players who used the tool showed reduced activity” rather than “the tool reduced activity.” Self-selection into tool use precludes causal claims.

5.5 Quality Ratings

Evidence quality was rated on a five-level scale with a direction indicator:

- **HIGH:** Randomised design with objective behavioural outcomes, extended follow-up, multiple operators or jurisdictions, validated measures, low attrition, and pre-registration; findings highly trustworthy
- **MODERATE-HIGH:** Strong designs meeting most but not all HIGH criteria; findings generally trustworthy with minor caveats
- **MODERATE:** Sound methodology with meaningful limitations; findings informative but require cautious interpretation
- **LOW-MODERATE:** Acceptable methodology with substantial limitations; limited confidence in causal claims
- **LOW:** Significant methodological concerns; findings may not be trustworthy

Direction indicators captured finding valence: positive (supports effectiveness), negative (ineffectiveness or harm), mixed (variable or conditional effects), or null (no significant effects). See 02.5 for full quality tier definitions and 02.4 for direction-value definitions; both include inclusion and exclusion criteria and worked examples.

5.6 Implementation Readiness Classification

Tools were classified into four readiness levels by evidence context. The glosses below are working summaries.

Table 02.2.4
Implementation Readiness Levels

Level	Definition
Deployed	Tool in active use at real operators (operator name in setting, “mandatory,” “operational,” industry-standard feature)
Validated	Tested with real gamblers in real settings (clinical trials, field trials on live platforms, community samples)

Level	Definition
Tested	Lab and research stage only (laboratory, simulated gambling, VR, survey-based message evaluation)
Conceptual	Framework, detection model, design principle, or theoretical (qualitative taxonomies, surveys of self-regulatory strategies, population-derived thresholds)

Classification described the tool's readiness, not the study's contribution. Attitudes and acceptability surveys, and laboratory evaluations of existing features, were classified as Conceptual. The same paper may yield different levels for different tools (e.g., an existing industry message = Deployed; an alternative being tested = Tested). Full definitions are in 02.5.

5.7 Implementation Notes Requirement

All extractions required Implementation_Notes addressing online translation potential, regardless of original study context. The field documented considerations for deploying the tool in digital gambling environments.

5.8 Dataset Storage Conventions

The 45 formal fields were stored in the deposited dataset across 37 columns using the following conventions.

Compound columns with sub-field labels. From v8, each sub-field was written as *Sub-field: CODE | free text*, with sub-fields separated by a period. The controlled-vocabulary code (where applicable) appeared before the pipe; free-text descriptors after. Sub-fields without a controlled vocabulary contained free text only. Five compound columns used this format:

- Country_Jurisdiction = Country + Regulatory_Context (free text in both sub-fields)
- Game_Type = Platform_Type + Money_Type (free text in both sub-fields)
- Delivery_Mechanism = Modality + Timing + Trigger + Frequency + Duration (controlled vocabulary on Modality, Timing, Trigger, and Frequency; Duration is free text)
- User_Interaction = Control + Engagement + Personalisation (controlled vocabulary on all three)
- Measurement_Method = Instruments + Timepoints + Procedure (free text in all sub-fields)

Example. A User_Interaction cell may read: *Control: VOLUNTARY | player-initiated through account settings. Engagement: INPUT-REQUIRED | numeric entry. Personalisation: USER-SET | player selects threshold value.*

Controlled vocabulary codes. Codes for Study_Design, Data_Source, Modality, Timing, Trigger, Frequency, Control, Engagement, and Personalisation are listed in summary form in Table 02.2.1. Full definitions, inclusion and exclusion criteria, and current record counts are in the Controlled_Vocabularies sheet of [OSF: 03_1_Taxonomy_Dataset.xlsx]. The Excel sheet is the authoritative source for code definitions; this codebook documents the structural framework.

Split column. Study_Sample is captured across three columns (Sample_N, Sample_Description, Sample_Severity) for granular coding; these count as one formal field.

Single-cell columns. The remaining formal fields occupy one cell per extraction.

An extractor-aid column (TL;DR) gave a plain-language summary of each record. It is not a formal field, does not count toward the 45-field total, and is not required for downstream analysis.

Note. RG = responsible gambling. Tool codes are defined in 02.3 Tool Codes and Domains. Quality tier rubrics are in 02.5 Evidence Quality and Readiness. Direction-value rubrics are in 02.4 Evidence Direction and Sample Size.